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广州市鑫国技术服务有限公司
检测专用章

REPORT ON FIRE PROPAGATION TEST TO BS 476 : PART 6 : 1989 ON "Polyurethane Foam (PU) of Double Skin Panel"

Client : Dongguan Tech Free Air Conditioning Mfy. Ltd.

Project : Fire Test on Building Materials & Structures Part 6

Client Ref. : --

Report No. : G20849FU206136

Date : 12 Aug 2020

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Appendix Specimen Photo

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1.0 Introduction

Fugro Technical Services (Guangzhou) Limited was commissioned by the client, Dongguan Tech Free Air Conditioning Mfy. Ltd. to conduct a fire propagation test of Polyurethane Foam (PU) of Double Skin Panel in accordance with **BS 476 : Part 6 : 1989 + A1 : 2009 Incorporating Corrigendum** No.1:2014.

The test was conducted by Fugro Technical Services (Guangzhou) Limited's approved sub-contract Laboratory.

The test samples were prepared by the client and received by Fugro Technical Services (Guangzhou) Limited on 16 July 2020 and given Lab. Code : FU206136.

2.0 Description of Samples

Six pieces of specimen, said to be "Polyurethane Foam (PU) of Double Skin Panel", each of nominal size of 225 mm x 225 mm were submitted. The overall bulk density and thickness of the specimen was found to be approximately 45±2kg/m^3 and 50 mm respectively.

Name and address of the sponsor	Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
Name and address of the manufacturer/supplier	Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
General description	Polyurethane Foam (PU) of Double Skin Panel
Material & composition for composite product	Galvanized Metal Skin, PU Inter-layer
Nominal thickness	50 mm
Nominal density	45±2 kg/m^3
Nominal mass per unit area	Nil
Colour	Silvery/White
Brand Name	TECH FREE
Model	RH2038101(P50GI)

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3.0 Purpose of Test

To determine the Index of Performance of the material when it is exposed to the conditions of the test specified in British Standard 476 : Part 6 : 1989 + A1 : 2009 Incorporating Corrigendum No.1:2014 "Method of test for fire propagation for products".

4.0 Test Procedure

Three specimens, backed with calcium silicate board, were tested with one face exposed to the specified heating conditions, in an apparatus conforming to paragraph 5 and illustrated in Figures 1 to 3 of the Standard.

The mean temperature rise above ambient obtained from three specimens is also shown in Figure 1 (i.e. with the actual calibration curve). The mean temperature readings for the material and the calibration curve were obtained at the following intervals from the start of the test : at 1/2 minute intervals up to 3 minutes, at 1 minute intervals from 4 to 10 minutes, and at 2 minutes intervals from 12 to 20 minutes.

From these readings, the index of performance for the material was determined as follows :

$$s_1 = \sum_{t=0.5}^{t=3} \frac{\theta_s - \theta_c}{10_t}, \quad s_2 = \sum_{t=4}^{t=10} \frac{\theta_s - \theta_c}{10_t},$$

$$s_3 = \sum_{t=12}^{t=20} \frac{\theta_s - \theta_c}{10_t}, \text{ and } S = s_1 + s_2 + s_3$$

Where S = Index of performance for each of the specimens tested and s_1, s_2 and s_3 are sub-indices

t = Time in minutes from the origin at which readings are taken.

θ_S = Temperature rise in deg. C for the specimen at time, t

θ_c = Temperature rise in deg. C for the calibration sheet at time, t

In computations only the positive value of $\frac{\theta_S - \theta_c}{10t}$ was used.

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5.0 Test Performing Date

23 July 2020 to 31 July 2020

6.0 Results of Test

The following test results were obtained for each specimen tested :

Specimen	Sub-Indices			Index of Performance
	s_1	s_2	s_3	S
A	0	0.32	0.09	0.41
B	0	0.41	0.10	0.51
C	0	0.36	0.10	0.46

7.0 Conclusion

The test results obtained for the sample tested are as follows :

Index of overall performance, I = 0.46
(Fire propagation index)

Sub-index, i_1 = 0

Sub-index, i_2 = 0.36

Sub-index, i_3 = 0.10

Note : If a suffix "R" is included in the above fire propagation index I, this indicates that the results should be treated with caution.

Remarks : The test results relate only to the behaviour of the test specimens of the product under the particular conditions of test ; they are not intended to be the sole criterion for assessing the potential fire hazard of the product in use.

Checked by : Glen

Approved by : Yan Cai Sheng (Deputy General Manager)

Date : 12/08/2020

Date : 12/08/2020

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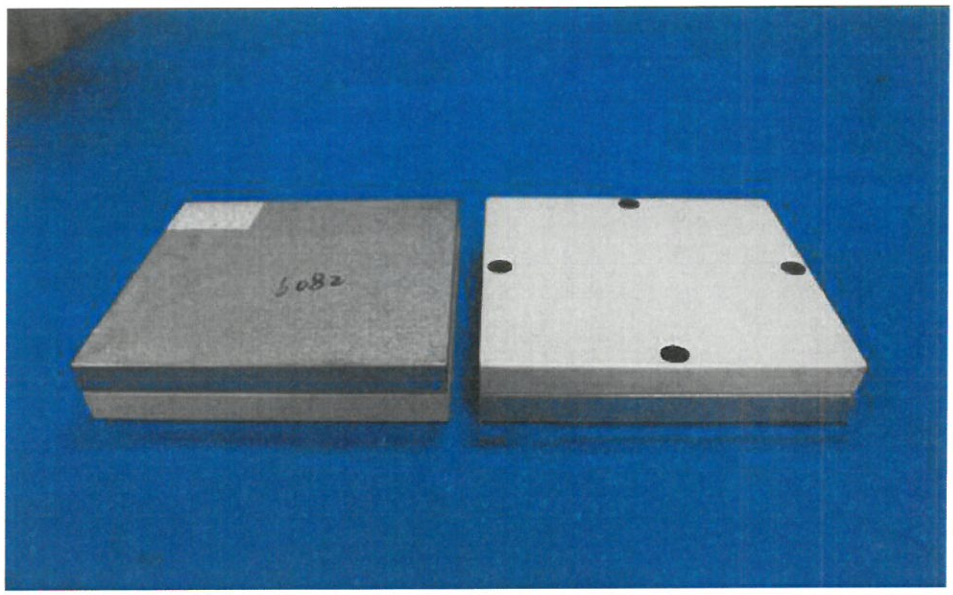

Sample ID: FU206136

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Client Ref. : --

Report No. : G20849FU206136(1)

REPORT ON CLASSIFICATION OF REACTION TO FIRE

Information Supplied by Client

Sponsor : Dongguan Tech Free Air Conditioning Mfy. Ltd.

Sponsor Address : No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.

Project : Fire Test on Building Materials & Structures Part 6 and Part 7

Product Description : Polyurethane Foam (PU) of Double Skin Panel

Test reports & test results in support of classification

Test Reports

Name of Laboratory	Name of sponsor	Test reports Nos.	Test method
Fugro Technical Services (Guangzhou) Limited	Dongguan Tech Free Air Conditioning Mfy. Ltd.	G20849FU206136	BS 476 Part 6:1989+A1:2009 Incorporating Corrigendum No.1:2014
Fugro Technical Services (Guangzhou) Limited	Dongguan Tech Free Air Conditioning Mfy. Ltd.	G20849FU206137	BS 476 Part 7:1997 Incorporating Corrigendum No.1:2014

Test Results

The following test results were obtained for each specimen tested:

Test method and test number	Results	Requirement of Class 0
BS 476 Part 6:1989+A1:2009 Incorporating Corrigendum No.1:2014	Fire Propagation index I=0.46	I≤12
	Sub-index i_1=0	i_1≤6
BS 476 Part 7:1997 Incorporating Corrigendum No.1:2014	Class 1	Compliance with Class 1

Classification

It is the opinion of this laboratory that, the tested sample complies with the requirement of Class 0 of UK. Building Regulations 2006 Approved Document B, appendix A paragraph 13.

Checked by: Eden Date: 12/08/2020 Approved Signatory: Date: 12/08/2020

Yan Cai Sheng (Deputy General Manager)

** End of Report **

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